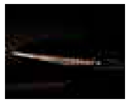
COVID-19 Spread To Animals
The virus that causes COVID-19 in humans can spread from humans to animals in some cases. **http://dgit.in/mar21-29**



Risk to the Middle Aged
Studies show that a middle-aged person has 100x greater risk of dying from COVID than a car accident. **http://dgit.in/mar21-30**

Fighting The pandemic

PART VII

The research directions to stop the spread of the novel coronavirus

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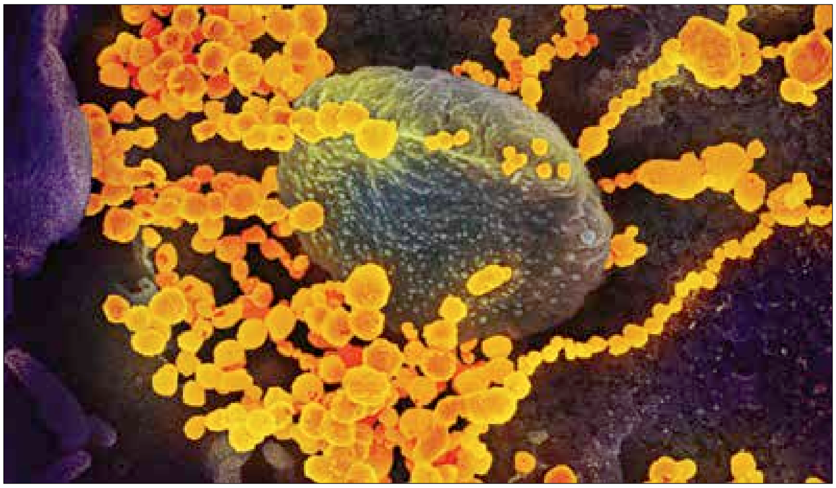


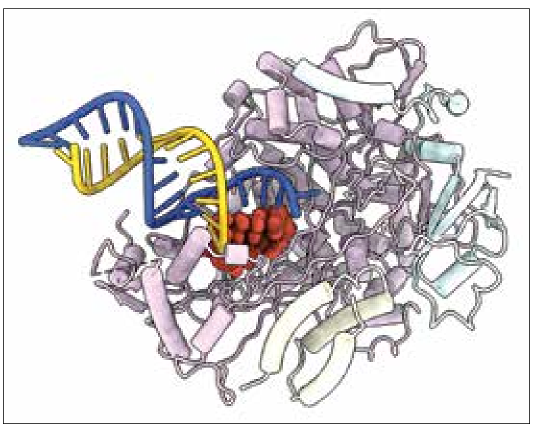
Credit: NIAID-RML.

SARS-CoV-2 viruses (the golden dots) emerging from the surface of cells cultured in a lab.

Between the onset of the pandemic and October 2020, there were an astonishing 87,000 scientific papers about the coronavirus. To put that number in perspective, nanoscale science has been getting unusual attention since the 1990s, and it took more than 19 years to accumulate 90,000 studies in the domain. The same amount of research was conducted on the coronavirus in a matter of five months. The research also showed that Chinese contribution to the science decreased drastically from around 47 percent of all papers between January to April 2020, but only contributed 16 percent from July to October 2020. Other countries where infection levels dropped, also saw a drop in research. The research showed that the number of papers increased, even though the number of authors of each paper reduced, and the amount of collaboration also reduced, which can be accounted for by the restrictions imposed during the outbreak. In Digit we have covered most of the interesting and important studies in this series of articles about scientists fighting the pandemic. Here are the latest updates.

REMDESIVIR

Researchers from the University of Texas have uncovered the exact process by which the antiviral drug remdesivir works against the coronavirus. The drug targets a part of the coronavirus that is used to replicate itself, preventing the process. Within the virus, the genetic information is copied for replication, much like the way a photocopying machine works. Remdesivir creates what is essentially a paper jam in the photocopying machine within the cell, which is an enzyme known as RNA polymerase. The researchers have identified a specific point where the block occurs, and what causes the block, and the researchers can make the blockage worse if they so desire. The understanding can allow researchers to create new antiviral drugs that are more potent in their action against the virus, but at the same time have fewer side effects for the humans. Currently, the five day regimen for



Credit: University of Texas at Austin.

A Remdesivir molecule, blocking a new RNA strand (yellow) from being replicated from the original RNA strand (blue), by the RNA polymerase in the coronavirus.